This file defines the communication contract between our banking client and banking server using Protocol Buffers.

Example: which services server provides, which remote function the client can call, input and output of each function how data should be structured and serialized etc.

banking.proto

```
1  syntax = "proto3";
2
3  package banking;
4
5  // Service for managing account balances.
6  service AccountManagementService {
7    rpc getBalance(BalanceRequest) returns (BalanceResponse);
8    rpc updateBalance(UpdateBalanceRequest) returns (UpdateBalanceResponse);
9  }
10
11 // Service for money transfers and transaction history.
12 service TransactionService {
13   rpc initiateTransfer(TransferRequest) returns (TransferResponse);
14   rpc getTransactionHistory(TransactionHistoryRequest) returns (TransactionHistoryResponse);
15 }
16
17 // Request to retrieve an account balance.
18 message BalanceRequest {
19   string user_id = 1;
20 }
21
22 // Response containing the account balance in paise.
23 message BalanceResponse {
24   string user_id = 1;
25   int64 balance_paise = 2;
26 }
27
28 // Request to update an account balance.
29 message UpdateBalanceRequest {
30   string user_id = 1;
31   int64 new_balance_paise = 2;
32 }
```

→ Specifies the file uses protocol buffer version 3

→ help organize our definition and avoid name conflict.

→ service: which is a group of related remote procedures.

remote procedure

→ retrieves current balance of a user.

→ updates user balance

another service.

→ To transfer money from one user to another.

→ check those transfer transactions

special proto buf field number. → message define structured data types that can be serialized and transmitted between client and server.

Same as above

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```
33
34  // Response after updating an account balance.
35  message UpdateBalanceResponse {
36    bool success = 1;
37    string message = 2;
38    int64 updated_balance_paise = 3;
39  }
40
41  // Request to transfer money between accounts.
42  message TransferRequest {
43    string from_user_id = 1;
44    string to_user_id = 2;
45    int64 amount_paise = 3;
46  }
47
48  // Response after a transfer attempt.
49  message TransferResponse {
50    bool success = 1;
51    string message = 2;
52    string transaction_id = 3;
53  }
54
55  // Request to retrieve a user's transaction history.
56  message TransactionHistoryRequest {
57    string user_id = 1;
58  }
59
60  // Details of an individual transaction.
61  message Transaction {
62    string transaction_id = 1;
63    string from_user_id = 2;
64    string to_user_id = 3;
65    int64 amount_paise = 4;
66    string timestamp = 5;
67  }
```

```
68
69  // Response containing a list of transactions.
70  message TransactionHistoryResponse {
71    repeated Transaction transactions = 1;
72  }
73
```

↳ repeated means field can contain zero or more message fields.

